Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/25/2007
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Project Title:	Reduced Gravity Studies of Soret Transport Effects in Liquid Fuel Combustion		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5294	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2008 per PI (10/07) NOTE: end date changed to 9/30/2007 per PI and GRC (10/06)		
Key Personnel Changes/Previous PI:	0		
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Task Description:	Soret transport, which is mass transport driven by thermal gradients, can be important in practical flames as well as laboratory flames by influencing transport of low molecular weight species (e.g, He, H and H2) and high molecular weight fuel species (e.g., C7H16) and O2. This research is concerned with investigating effects of Soret transport on combustion of liquid fuel droplets. Experiments will involve normal-gravity and reduced-gravity experiments on combustion of liquid fuel droplets in environments where Soret effects significantly influence transport of fuel and oxygen to flame zones. The experiments will employ pressures ranging from 0.3 to 10 atm, with extensions to 30 atm also being considered. Experiments will also be performed where Soret effects are not expected to be important. Droplets initially in the 0.5 to 1 mm size range will be burned. Data will be obtained on influences of Soret transport on combustion characteristics (e.g., droplet burning rates, droplet lifetimes, gas-phase extinction, and transient flame behaviors) under simplified geometrical conditions that are most amenable to theoretical modeling (i.e., spherical symmetry). The experimental results will be compared with existing theoretical models as well as new models that will be developed as part of this program. The knowledge gained will be of use in evaluating combustion mechanisms under reduced gravity, which is of interest for spacecraft fire safety. It is noted that in fall 2005, the PI became involved with a different program that includes a team of investigators from various universities as well as from NASA. This program, which is termed the Flammability and Extinction Experiment (FLEX), is focused upon using droplets to study flammability and extinction under reduced-gravity conditions of relevance to spaceflight and human exploration of space. The FLEX program is both experimental and theoretical in nature. Experiments presently include reduced-gravity studies as well as normal-gravity comparison experiments. In the research, individual droplets are dispensed, deployed and ignited in reduced gravity (as well as normal gravity). Theoretical approaches include developing computational and analytical models, with a particular emphasis on evaluating influences of gas-phase diffusive transport as well as droplet internal flows, which may arise from Marangoni stresses as well as droplet deployment and ignition. The present research on influences of Soret transport on droplet combustion is strongly complementary to certain aspects of the FLEX program, e.g., research on influences of gas-phase transport properties on flammability and extinction of droplets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research has the potential to produce new and useful information on influences of Soret transport combustion mechanisms, both in normal gravity as well as reduced gravity. For example, Soret transport can be important in low-gravity flames by significantly influencing transport of O2 to a flame zone when low molecular weight (e.g., He) or high molecular weight (e.g., SF6) species are present in the gas phase. Low gravity combustion characteristics such as flame sizes, flame temperatures and sooting can be significantly influenced by Soret transport. Increasing knowledge about these effects is important for applications such as spacecraft fire safety. Soret transport can also influence transport of high molecular weight fuel species, which are commonly employed in terrestrial engines. The present research has the potential to provide new information that will increase understanding of combustion mechanisms that are relevant to terrestrial applications that involve use of liquid fuels.
Task Progress:	Work has principally involved performing normal-gravity and reduced-gravity experiments, analyzing data from these experiments and performing theoretical analyses. Summaries of the progress are given below. Normal-Gravity and Reduced Gravity Experiments Reduced gravity and normal gravity combustion experiments were performed with individual fiber-supported droplets that were initially about 1 mm in diameter. These experiments were performed with a goal of investigating influences of Soret transport and multicomponent diffusion on droplet flammability characteristics. Droplets composed of heptane, methanol or propanol were investigated and a thermocouple and radiometer were employed in conjunction with video cameras to obtain data on flammability behavior. The environment was composed of an air-inert mixture at 0.1 MPa and about 298 K, where helium, carbon dioxide and xenon were separately used as inerts. In the experiments, the amount of inert gas in the environment was progressively increased until droplets could no longer be ignited. Droplet ignitability depended on the gravitational level as well as the inert gas in the environment. In general, the amount of inert gas required to suppress combustion was higher in reduced gravity than normal gravity. Use of xenon required significantly higher mole fractions for suppression of combustion relative to helium or carbon dioxide. The data indicate that ambient gas transport properties play an important role in determining flammability of droplets. For example, Soret diffusion likely promotes transport of O2 to flame zones when Xe is used, delaying extinction. Conversely, use of He as an inert will likely promote extinction, with Soret transport playing a significant role by inhibiting transport of O2 to flame zones. Analysis of the experimental data has also indicated that droplet burning rates depend significantly on the ambient gas conditions as well as the gravitational level. Theoretical Modeling Efforts were directed towards development of theoretical models to predict the influences of Soret transport on combustion behavior. These efforts involved computational modeling of diffusion flame behaviors for the opposed-flow configuration, which is a one-dimensional situation (with planar or cylindrical symmetry), where use was made of CHEMKIN software. Calculations were performed for various fuel and oxidizer combinations with chemical kinetic mechanisms of varying complexity. Some calculations included Soret transport while others did not. Results indicate that Soret transport can play an important role for these opposed-flow configurations, but that the influences of Soret transport may be weaker than for the spherically-symmetric case that corresponds to droplet combustion in reduced gravity. An important variable in the opposed-flow calculations is the strain rate, and the calculations have suggested that Soret transport may be most important at the low strain rates that would likely prevail in low-gravity situations with low-speed flows.
Bibliography Type:	Description: (Last Updated: 04/29/2010)
Papers from Meeting Proceedings	Wei JB, Shaw BD. "Reduced Gravity Combustion of Propanol Droplets with Various Inert Gases." Presented at the 2005 Fall Meeting of the Western States Section of the Combustion Institute, Stanford University, Stanford, CA, October 17-18, 2005. 2005 Fall Meeting of the Western States Section of the Combustion Institute, October 2005, Paper 05F-54. , Oct-2005